

COMBUSTION OF CLEAN BIOMASS AT HIGH STEAM PARAMETERS OF 540°C RESULTS FROM A NEW 120 MW_{TH} UNIT

M. Bolhàr-Nordenkamp, A. Wojcik, F. Gartnar, I. Tschanun, S. Kaiser
Austrian Energy & Environment GmbH & Co KG; Siemensstrasse 89, A-1210 Vienna
Phone: +43/1/250 45 - 0; Fax: +43/1/25 84 139; Mail: markus.bolhar@aee-group.com

ABSTRACT: Biomass fuel exists in various forms, traditionally as wood, bark, harvesting residues sewage sludge and organic waste resulting from agricultural industry. With the actual on or near site availability of biomass fuels and the trend for independent power production the demand for reliable biomass fired power plants is increasing. To satisfy this demand, a combustion technology, which can cope with a broad fuel range as well as with low ash melting points, and is able to produce steam at high steam parameters without the risk of corrosion on the other hand, is necessary. To comply with these demands the standard bubbling fluidized bed technology normally operating in combustion mode was modified to a staged combustion concept. The first FBC-plant with such high steam parameters was successfully put in operation last year in Umeå/Sweden. The boiler, which is designed for the combustion of a mixture of wood fuels, peat Salix and energy crops, has a thermal capacity of 120 MW_{th} which equals 37 MW electrical output. The plant can be operated between 100% and 35% load. The produced live steam at 143 bar_a and 543 °C is not only used for electric power production but also for district heat generation. The whole plant consists of an integrated fluidized bed boiler, an electrostatic precipitator and a SCR-system for flue gas cleaning.

Keywords: fluidized bed, combustion, industrial scale application, dedicated biopower plant.

1 INTRODUCTION

As a worldwide supplier of fluidized bed combustion plants the reference list of Austrian Energy & Environment comprises over 90 steam generators with capacities from 9 t/h to 400 t/h. with a fuel range from coal, biomass, sludge, RDF to various industrial wastes.

Depending on special requirements of the various fuels and the size of the boiler different combustion systems like the bubbling fluidized bed or an external circulating fluidized bed technology will be used.

All of these systems have been developed by AE&E and will allow maximum flexibility in the range of fuels. Beside environmentally sound combustion with low emissions, maximum boiler efficiency is the main target.

In this paper operating experience of a new biomass fired plant with high steam parameters in Umeå/Sweden are presented.

2 COMBUSTION TECHNOLOGY

The applied combustion technology in biomass combustion plants is the **ECOFLUID®** bubbling fluidised bed. The main feature of this technology is the principle of staged combustion of the fuel. The oxygen level in the fluidised bed is limited and hence only a part of the fuel is combusted, whereas the rest of the fuel is gasified. This can be achieved by adding a substoichiometric amount of oxygen (λ approx. 0.35) to the fuel. However, in order to keep a constant λ and temperature in the bed, this would result in a fluctuation of the fluidisation air volume flow and hence in fluidisation of the bed in accordance to the heating value of the fuel. Since this effect is not desired, the primary air is mixed with recirculated flue gas. This allows the control of λ and the bed temperature as well as keeping the fluidisation of the bed at a constant level.

This substoichiometric bed operation allows the control of the bed temperature in the range between 650°C-820°C. Therefore, also fuel with low ash melting temperature can be burned without any sintering problems in the bed. The standard operation temperature

of the fluidised bed is approximately 760°C.

The gasification gases rising from the bed are fully combusted by adding secondary air to the boiler. This causes a rise in temperature and oxygen content, as seen in Fig. 1. The turbulence in this area of the first pass results in very low CO-values in the fuel gas.

For high calorific fuels recirculation gas is injected above the secondary air to “cool” the flue gases in the post combustion chamber by stretching the combustion process. This staged combustion concept results in a homogenous and moderate temperature profile in the furnace and first pass of the boiler and thus low NO_x emission. If needed, NO_x emissions can be easily controlled by installing a SNCR at the appropriate temperature level in the boiler. By using refractory lined superheaters in the first pass and the second pass corrosion problems can be minimized although high steam parameters can be obtained. These properties enable the **ECOFLUID®** bubbling fluidised bed to handle a broad fuel range with different heating values as well as corrosive fuels (e.g. net calorific values in a range of 3 to 20 MJ/kg) [1-4].

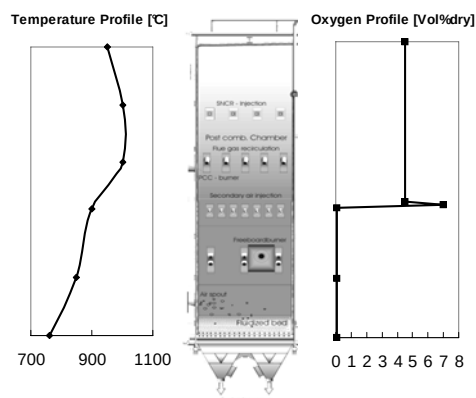


Figure 1: Temperature and oxygen profile in the combustion chamber of the fluidised bed boiler

3 BIOMASS FIRED POWER PLANT UMEÅ/ SWEDEN

The technical concept comprises a biomass combined heat and power plant (CHP) with a new bubbling fluidised bed boiler. With an estimated operation of 8000 h/year, 299 GWh of electricity and 562 GWh of district heat can be produced by burning 144.000 up to 275.000 tons (depending of LHV) of solid fuel. The investment volume was nearly 100 Mio Euro.

The fuel consists of following fuel types:

- Wood fuels such as forestry residues (branches, tops), wood chips, bark, sawdust, sawmill and carpentry waste
- Peat
- Salix and energy crops

The permit was granted in the beginning of 2007, the erection was started at the end of 2007 and the boiler reached the PAC in January 2010. Reason for the late PAC was the planned shut-down of the boiler during the warm months of the year between June and September since the boiler is operated by the district heating demand of the town.

The plant consists of the following main subsystems:

- Bubbling Fluidising Bed boiler (BFB); Type ECOFLUID
- Cleaning system for boiler heating surfaces (sootblower system)
- External fuel system (not in scope of AE&E)
- Solid fuel intermediate storage, conveying, metering and feeding equipment
- Start up and support burners for light fuel oil
- Combustion air and Flue Gas Recirculation (FGR) system
- Induced Draft (ID) fan (not in scope of AE&E)
- Sand storage and dosing
- Bed ash extraction, coarse particle separation and conveying system
- Boiler ash extraction and conveying system
- Ash cooling water system
- Electrostatic precipitator for fly ash removal from flue gas and fly ash conveying system (not in scope of AE&E)
- Ash storage silo (not in scope of AE&E)
- Selective Catalytic Reduction (SCR) of nitrogen oxides (not in scope of AE&E)
- Scrubber (not in scope of AE&E)
- Boiler water-steam system (incl. make up water, feedwater system etc.)
- Feedwater dosing system (NaOH and NH₄OH)
- Water and steam sampling system
- Heat transfer system
- Low pressure (LP) steam system
- High pressure (HP) steam system
- District heating system (not in scope of AE&E)
- Condensate system
- Condensate treatment system (Condensate Polishing Plant - CPP)
- Service systems (pressurised air, cooling water, lube oil system)
- Electric, control and protection systems

3.1 Basis for the design of the fluidized bed combustion
Considering fuel properties and range of heating values, the decision was made to use an **ECOFLUID®** boiler type with substoichiometric combustion in the bed and a secondary air supply for the post combustion chamber.

Table I: Overview of the design values

Design Values	
Max. fuel heat rate	123,3 MW
Lower heating Value	6.000 ÷ 12.000 kJ/kg
Moisture	30 ÷ 58 %
Ash content	max. 4,2 %
Fuel flow	max. 69,0 t/h
Control range	35 ÷ 100 %
Combustion temperatures	700 ÷ 920 °C

The following emissions are 24 h average values related to 6 % O₂ content in dry flue gas.

Table II: Overview of the emission values

Pollutants according decree	Emission limit
Carbon monoxide CO	30 mg/Nm ³
Sulphur dioxide SO ₂	40 mg/Nm ³
Nitrogen oxides NO _x as NO ₂	70 mg/Nm ³
Dust	5 mg/Nm ³
Hydro chloride HCl	10 mg/Nm ³
Hydro fluoride HF	1 mg/Nm ³
Mercury Hg	0.05 mg/Nm ³
Dioxins and Furans PCDD/F	0.05 ng/Nm ³

The emission values for CO and NO_x have to be obtained with the fluidized bed combustion respectively with the SCR. The emission value for Dust has to be obtained with the installed ESP system.

Together with the wood wastes incombustible parts like stones, nails, screws and metals are usually fed to the fluidized bed, where they collect and hinder the fluidization. To prevent this, an open type air distributor in AE&E –design was built. This air distributor enables the drain of bed material over the whole bed cross section during operation.

To maintain a constant fluidization of the bed at a constant bed temperature recirculated flue gas is added to the primary air in direction to the bed. To control the temperature in the post combustion chamber recirculated flue gas is used, too.

3.2 Plant description

Fluidized Bed Combustion and Steam Generator

The steam generator is designed for the following data:

Table III: Steam generator design data

Live steam flow	155 t/h
Live steam pressure	143 bar(a)
Live steam temperature	543°C
Guaranteed flue gas temperature	max. 300°C (outlet 4 th pass) max. 200°C (outlet 5 th pass)
Feedwater temperature	140 °C

The steam generator (see Figure 2) is a top supported 3-pass boiler with natural circulation, where the passes are integrated in the evaporator membrane-walls, except the rear- and side walls of the 3rd pass, which operate as superheater 1. The bubbling fluidized bed is situated in the lower part of the 1st pass. The membrane-walls in the fluidized bed and a part of the post-combustion chamber and 2nd pass are covered with refractory material due to erosion protection and thermal reasons. The primary air enters the bubbling fluidized bed via the open type air distributor.

In the lower third of the combustion chamber, the cross section is reduced to enhance the mixing of the gasification gases and the secondary combustion air. The secondary air is distributed over this reduced cross section together with the recirculated flue gas for temperature control in the post combustion chamber. This results in a staged combustion and a temperature increase to 870°C up to 950°C at maximum in the post combustion chamber.

The rear- and side membrane walls of the 3rd pass operate as superheater 1. Two superheaters (SH2, SH3) are situated in the third pass, the last superheater (SH4) is situated in the 2nd pass and is designed as wing walls, to reduce erosion and corrosion. To reach the high live steam temperature of 543°C, all superheaters in the 2nd and 3rd pass are arranged in counter-current flow. The fourth pass, built as carbon steel casing, contains the 2nd part of the economizer, which cools the flue gases down to 280°C. The ESP and SCR is installed downstream the 4th pass. After the SCR the flue gas is directed into the 5th pass, which contains the 1st part of the economizer and the heat transfer system and cools the flue gas down to 150°C.

Heat Transfer System

The Heat Transfer System has the function to cool down the flue gas at the outlet of the 5th pass and to transfer the low temperature heat to the preheaters for primary and secondary air. The flue gas is cooled in a tube heat exchanger with circulating water. The hot water heats the combustion air with finned tube registers.

Solid Fuel System (internal)

The solid fuel is supplied by a belt conveyor into a fuel distribution chute inside of the boiler house and is split to two surge bins by two speed controlled fuel screws. In the surge bins a minimum level must be maintained to enable proper working of the rotating screw reclaimers.

The fuel discharged from the surge bins is conveyed to two metering bins at the boiler. Out of each bin two metering screws convey the fuel into the furnace according to their rotation speed. The rotation speed determines the fuel heat input into the furnace and is used for the calculation of the necessary combustion air demand. During normal operation all four metering screws are in operation with equal rotation speed. In both fuel metering chutes a minimum level of solid fuel must be maintained to prevent leakage air entering the furnace via the fuel system. A pneumatic actuated slide gate valve isolates the fuel supply system from the combustion chamber in case of emergency or if the conditions for solid fuel combustion are not reached. The fuel is distributed into the bed by means of four air spouts.

Burner System

Start up of the fluidized bed boiler is carried out with light-oil burners, heating the bed material to the ignition temperature of the wood. Two burners are installed in the freeboard of the bubbling-fluidized bed, another two in the post combustion chamber to ensure the desired emission limits by maintaining the 850°C for two seconds during start-up and also for fuels with very low heating value.

Bed Material and Ash Handling

The ash handling system consists of 2 different systems:

Bed ash system: The ash of the combustion chamber (1st pass) is handled in this system. The bed material, ash and coarse particles are drawn off at the bottom of the furnace via three cooling screw conveyors. With the help of drum screens, which are integrated in the screws, the coarse particles are separated from the fine particles. Fine particles can be conveyed pneumatically back into the furnace in order to reduce sand consumption of the boiler or can be sent to the ash silo for disposal. The separated

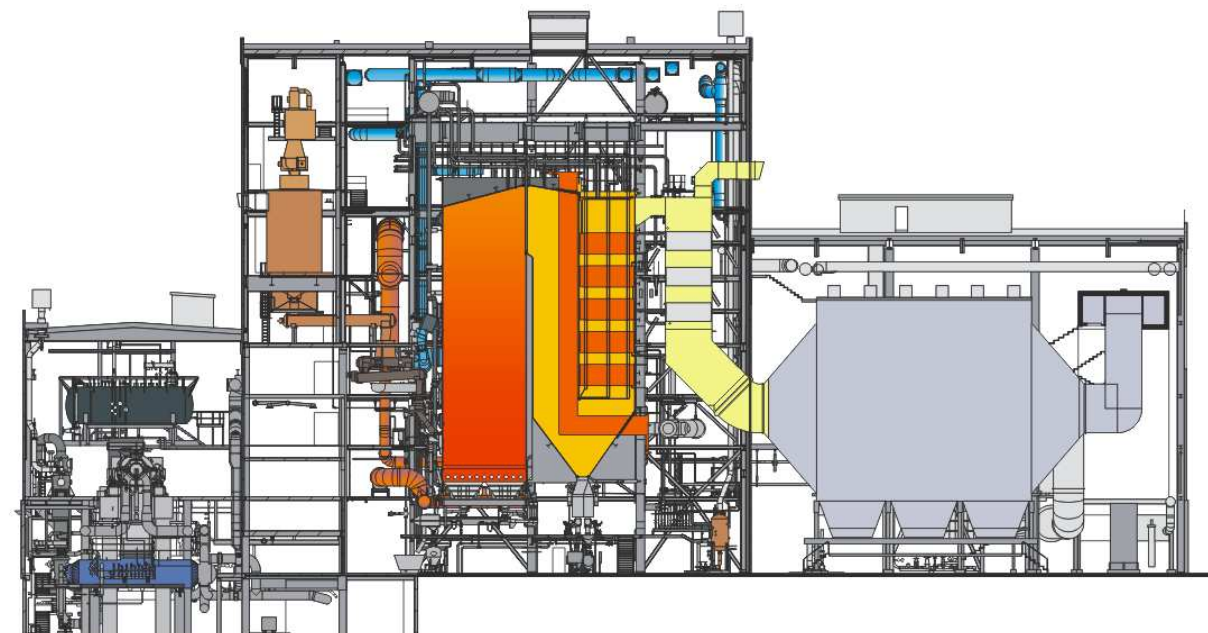


Figure 2: Cross section of the biomass power plant

coarse particles are collected in containers.

Boiler ash system: In this system the ash from 2nd and 3rd pass is collected by two cooling screws and stored in intermediate bins. The ash is then conveyed pneumatically to the ash silo.

Cleaning of the Heating Surfaces - Sootblower System

For cleaning of the heating surfaces in the boiler 19 soot blowers supplied by 3 steam lines are installed. The steam for the soot blowers is extracted from the inlet of the superheater 3 and the pressure is reduced to the necessary operation pressure of the soot blowers.

4 OPERATION RESULTS

The commissioning started in October 2008 and the plant was handed over to the client after 6 weeks of successful trial run in January 2010. Reason for the late PAC was the planned shut-down of the boiler during the warm months of the year between June and September.

Some delays occurred during commissioning because of the strong winter. Actually the turbine (not in scope of AE&E) was not delivered on schedule because the cargo ship was frozen in the river Elbe.

The distribution of the solid fuel in the daily bins was not satisfying because of the usage of rotating distribution chutes on top of the bins and a wrong dropping angle of the solid fuel. As a result a subsidence slope of approx. $\pm 60\%$ occurred inside of the bins especially when operating at high levels. Besides that the weight of the solid fuel stored in the bin was too much (at high levels) for the screw reclaimer to operate. To solve the problem the normal operation level was set down to 40%, which led to a safe and satisfying continuous operation.

The installation of the heat transfer system instead of air/flue gas heat exchangers was a great success. After solving some minor problems with the pressure maintenance of the system, the cooling of the flue gas at outlet of the 5th pass and the associated preheating of the primary and secondary air was done successful within the whole operation range.

The design of the **ECOFLUID**[®] fluidized bed combustion was confirmed during commissioning and the following commercial operation by the achieved operation data. All emission values are far below the guaranteed limits. The furnace could be operated easily within the defined large fuel range. Even at loads of only 35% a stable, safe and automatic controlled operation was possible.

Finally, it can be stated that this power plant especially considering the high steam parameters is a big step forward in the green energy production from Biomass.

5 NEW COMPACT BOILER CONCEPTS FOR HIGH STEAM PARAMETERS

The aim of the development was to include the operation results from various biomass plants and incorporate these results into a new design, which offer benefits in the following points.

- Weight reduction of the boiler pressure parts which offers the possibility to use a bottom supported boiler type even for higher steam

load outputs

- Focus on low NO_x emission by optimization of the combustion conditions
- Focus on low CO emissions by optimization of the combustion conditions and sootblower arrangement
- Maintaining and improving the high availability approach of the concept

The following graphics shows the boiler layout of such a concept whereupon the different characteristics shall be discussed.

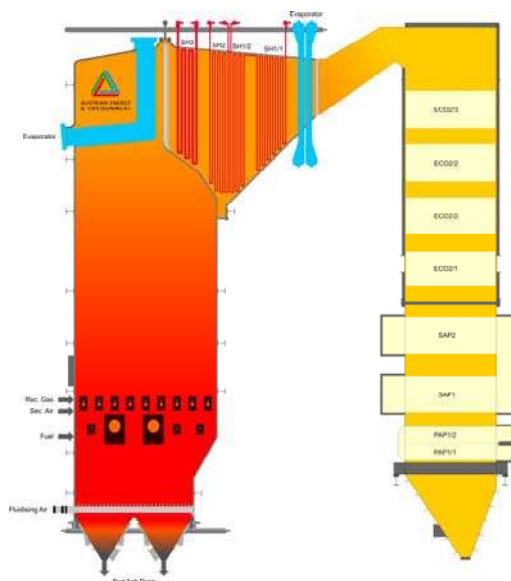


Figure 3: Layout of the ECOFLUID-CC-Concept for biomass

Using this boiler design steam loads up to 150 t/h can be constructed with a bottom supported design with the potential for even higher steam loads in the future. The flue gas inlet temperature to the first superheater can be adjusted by evaporator bundles if necessary in order to adapt to different corrosion potentials in the fuel. The arrangement of the superheaters is done in a way that the superheater design has no influence on the furnace design. No superheaters are placed directly in the furnace which allows the usage of less costly materials for the superheaters. If it is necessary to increase the superheater surface in order to adapt the live steam temperature this can be done by extending the crossover duct without interference of the furnace design. The flue gas velocity in the superheater bundles can be adapted by a variation of the superheater tube spacing. Beneficial is also that a maximum furnace volume can be achieved at a given boiler height.

Since there will be always ash entrainment from the fluidized bed into the superheater ducts, the horizontal pass was designed in a hopper form. With this solution entrained fly ash is separated by gravity in the crossover duct and can be extracted before the economizer path.

Another positive aspect of the decoupled superheater arrangement is that the influence of sootblowing on the furnace is minimized. Impulses of sootblowing can be up to ten times higher than the flue gas flow and can have a significant influence on the combustion process and can result in high CO peaks.

Using this design clean biomass can be combusted with high steam temperatures in a flexible compact bottom supported boiler design, integrating the special requirements of the fuel biomass together with a cost efficient design.

6 CONCLUSION

The **ECOFLUID**[®] technology is capable to combust biomass fuels within a broad operation range securely at high steam parameters. Low stable part load operation can be achieved as well as high superheating temperatures of 540°C without the risk of corrosion. The operating experiences from the plant in Umeå were the basis for the further development of the technology. The new boiler concept comprises a compact solution which provides the basis to convert biogenic fuels to steam and finally electricity with the benefit of reduces investment costs. This is a major contribution to enhance green power production on an environmentally friendly basis.

7 REFERENCES

- [1] Tschanun, I., Holarek, M., Gartnar, F., Glatzer, A. (2001): Westfield, Fife/Scotland becomes World's first to burn Poultry Litter in FBC, Proc. of the Power Gen Europe 2001, 29-31. May 2001, Brussels, Belgium.
- [2] Bolhàr-Nordenkamp, M., Gartnar, F., Tschanun, I., Kaiser, S. (2005): Operating Experiences from FBC-Plants using Various Biomass Fuels, Proc. of the 14th Bioenergy Conference, 17-21 Oktober 2005, Paris, France.
- [3] Tschanun, I., Mineur, M., (2003): Biomass Combustion with State of the Art Bubbling Bed Steam Generators, Proc. of the Power Gen Europe 2003, 6-8 May 2003, Düsseldorf, Germany.
- [4] Tschanun, I., Franz, P., Gartnar, F. (1998): Klärschlammverbrennung bei der Vera GmbH – Hamburg mit dem Wirbelschichtverfahren der Austrian Energy, Proc. of the 1st Berliner Klärschlammtagung “Integrierte Klärschlamm-entsorgung”, 25 – 27 May 1998, Berlin, Germany.